

An Analysis of Clinical Variables and Gene Expression Data across Biopsies from Patients with Lung Adenocarcinomata

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Introduction

Loading and Cleaning of Annotation and Expression Data

We want to clean up our annotation file. To do this we must first take a look at the file.

We can see here, that there are indeed values missing in a lot of the columns. We can also see, that the first three segments = or the first 12 digits = of the barcode encode the patient ID. We thus replace `paper_patient` with the barcode trimmed to position 12.

Since we only want the barcode and paper columns, we discard anything not matching to that selection. This leaves us with 20 paper columns.

As this also makes the `'paper_'` prefix superfluous we strip it from the paper columns. Furthermore, we noticed textual NAs, which were entered into the data set as `'[unknown]'`, `'[not available]'` and `'[Not Available]'`. We thus strip these and replace them by real NAs. We perform the same operation on the expression data, whilst we're at it. The `paper_Tumor.stage` column encodes the Ajcc pathological stage. Thus, we replaced it with the ajcc staging.

When looking at the counts of NAs, the expression data are complete, whilst the annotation data have 5623 NAs in total distributed relatively evenly across all columns, with patient and Tumor.stage being the exceptions.

From the head-call above, we know that all of the text columns are characters, even when a factor or numeric type would be more appropriate. We thus go through the cleaned annotation table and factorise, where appropriate. Furthermore, where factorisation is inappropriate, we try to coerce each column to numeric type (accepting both comma and period as decimal delimiters) and discard all failed coercions. Lastly, we double check, that we see the intuitively appropriate classes for each column.

We know from the documentation of the data (<https://gdc.cancer.gov/resources-tcga-users/tcga-code-tables/sample-type-codes>), that the leading numbers of the fourth segment tell us whether the sample is a tumour or not. We use this to encode a new column entitled `'is.tumour'` storing that information. Since this segment only has the values of `'01'` and `'11'`, and since `'11'` codes for healthy tissue and `'01'` for tumour tissue, a boolean suffices.

With our annotation data (and incidentally also expression data) cleaned up and usable, we move on to the next step.

Data Exploration

Clinical Annotation

Pathologic stages 1A and 1B are the most common among the patients.

Patients are between 41 and 85 years old, with a mean age of 65.5 years.

There are 110 female and 73 male patients.

Most of the patients have been smokers or are smokers, only a small proportion (14%) are life long non-smokers.

EML4-ALK and CLTC-ROS1 are the most common fusions, CLTC-ROS1 is present in the prox.-inflam expression subtype and EML4-ALK is present in the TRU subtype

Most high CIMP methylation signature samples are in Cluster Group 3 and 4, in Cluster Group 1 are only low CIMP methylation samples.

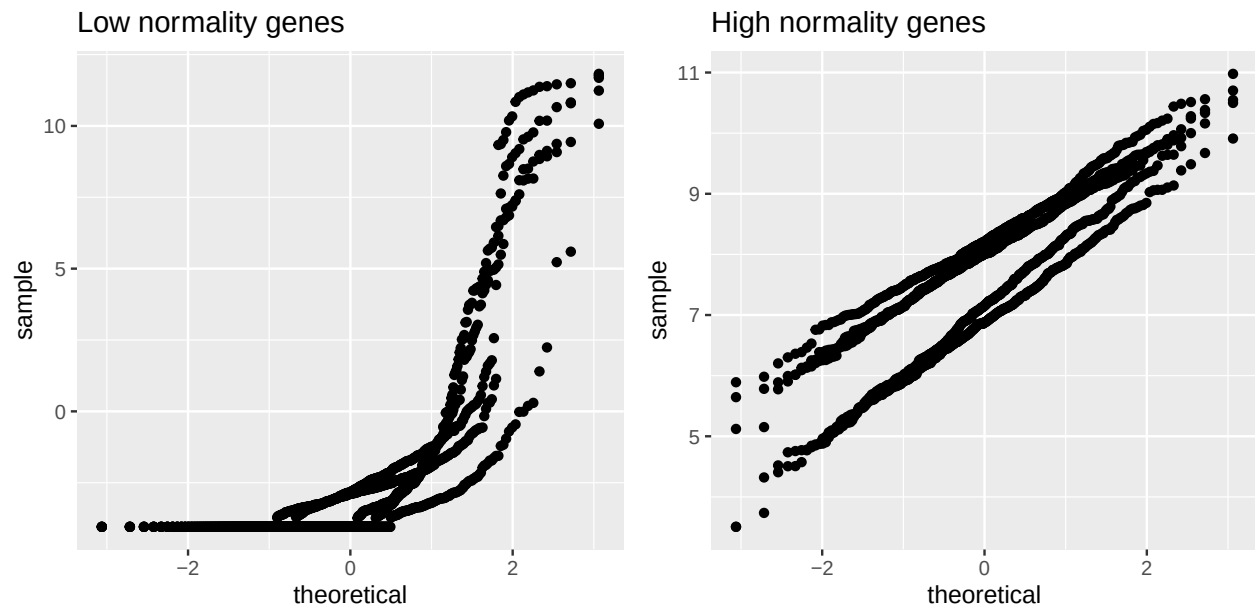
cross tabulation of iCluster Group with Sex shows that the male and female patient samples are not equally grouped, the biggest subgroup is female patients with iClusterGroup 5

cross tabulation of iCluster Group with Age shows no clear association

Expression Data

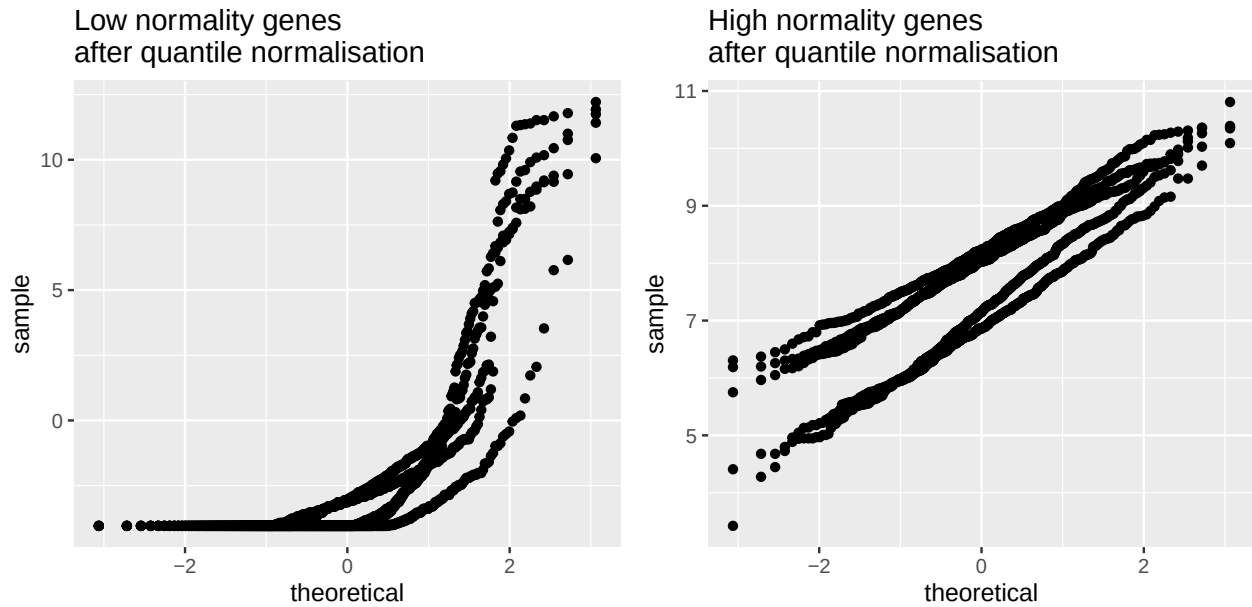
We first aim to assess the normality of the gene expression data. As we have 3000 different genes, analysing a histogram or QQ-plot for each one is not feasible. Instead we perform Shapiro-Wilks tests for each gene across the 455 samples. We can then plot a histogram of p-values to roughly ascertain the normality across different genes and we can calculate that only ~82% of genes gave a p-value lower than 0.05.

For better insight we select some genes with high and some with low p-values and make QQ-plots on these.



Next, we perform quantile normalisation of the gene expression data. The Shapiro-Wilks test now yields only ~77% of genes with a p-value lower than 0.05.

We can redo the QQ-plots for the same genes to see what changed. The distributions look similar, but Shapiro-Wilks testing shows generally lower normality. Because of this, we utilise non-quantile-normalised expression values



Data Analysis

Dimensionality Reduction of Annotation Data

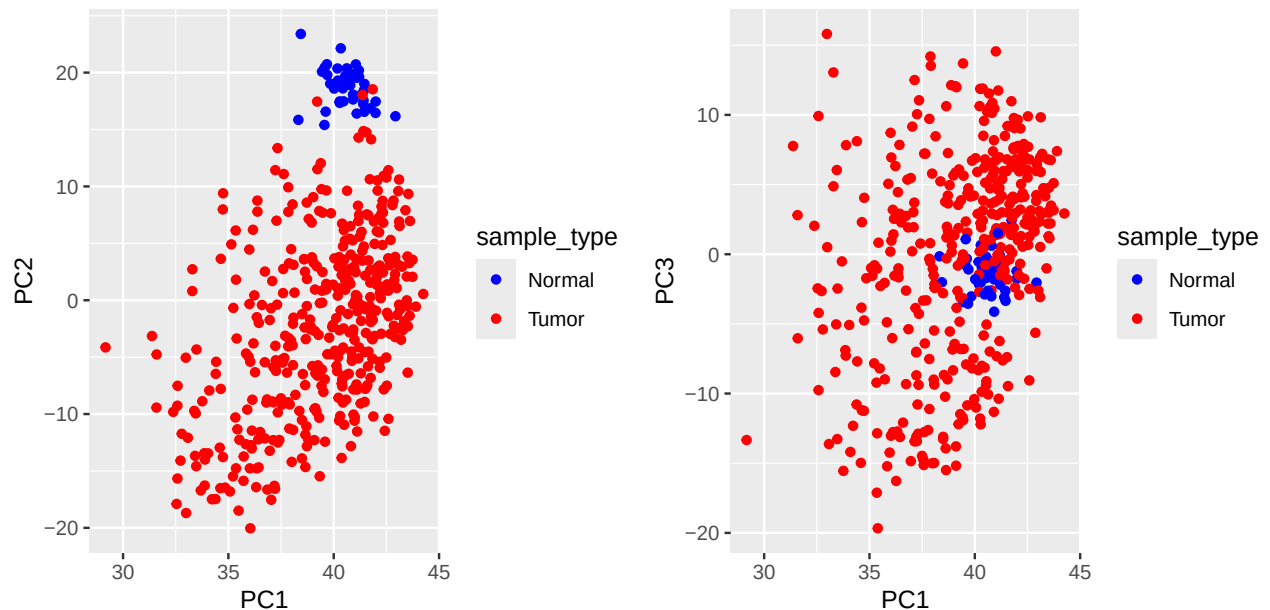
R Markdown

The expression data is first presented in a matrix in which the rows are named after the genes.

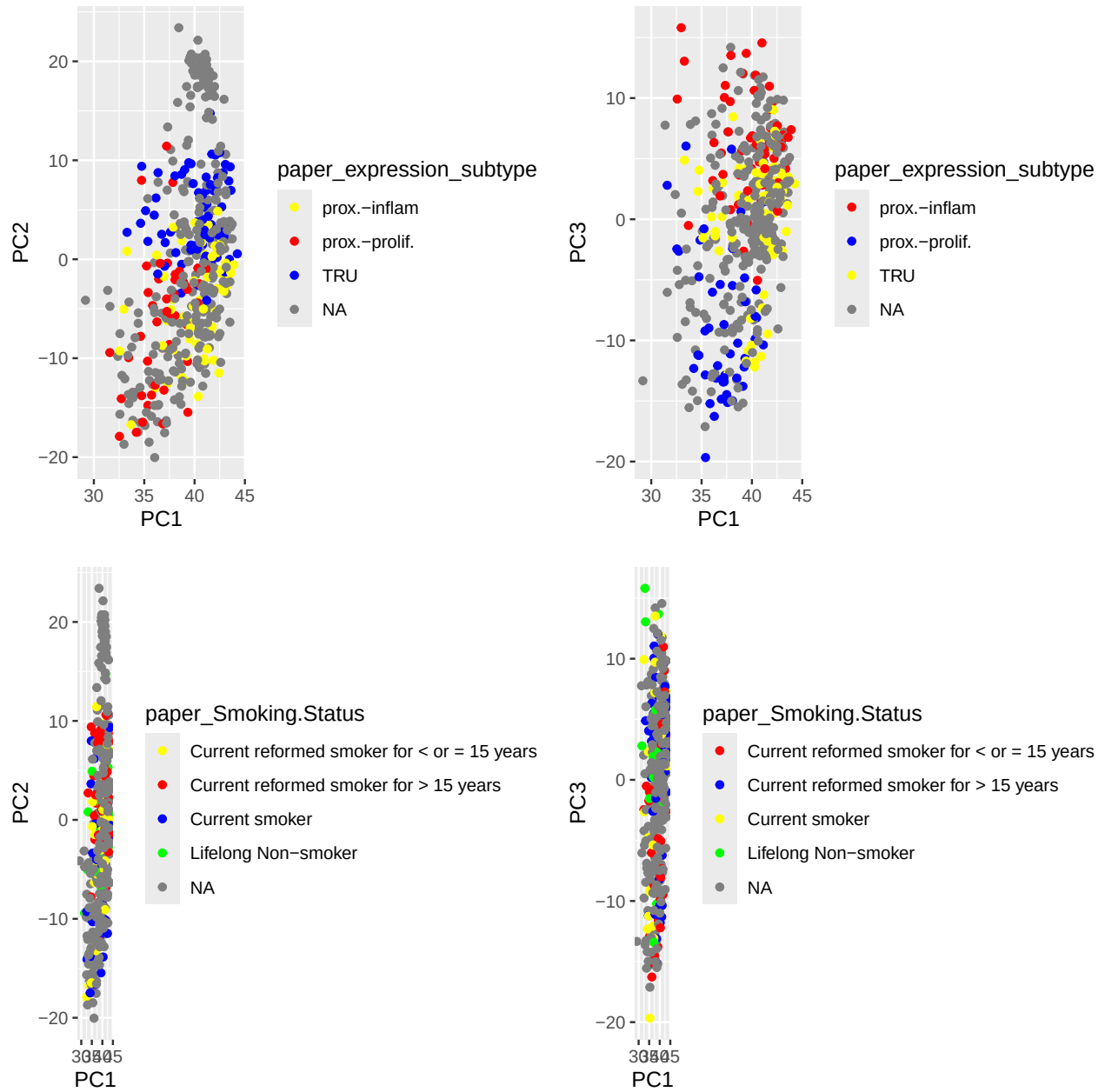
The variance is calculated for each gene and represented as a vector.

We can now perform a principle component analysis. We need to transpose the data, because the PCA function is expecting genes as columns and samples as rows. we only choose the genes with high Variance.

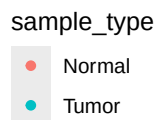
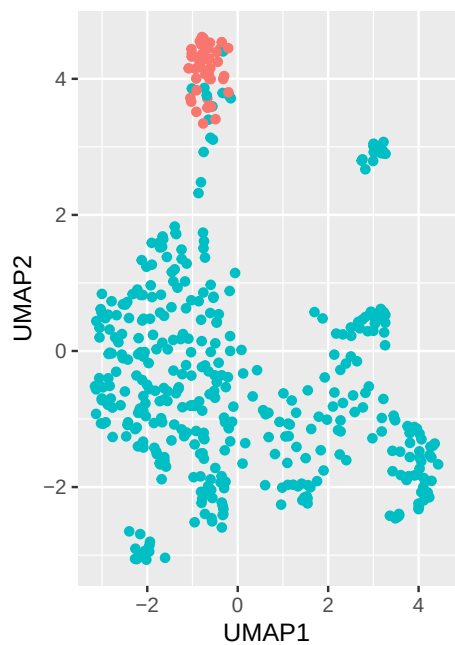
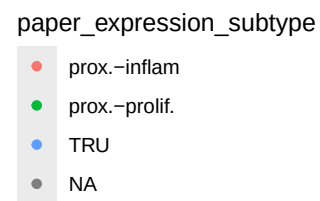
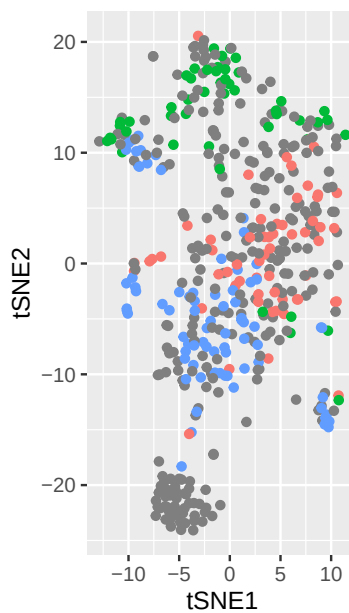
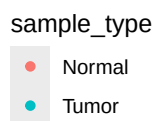
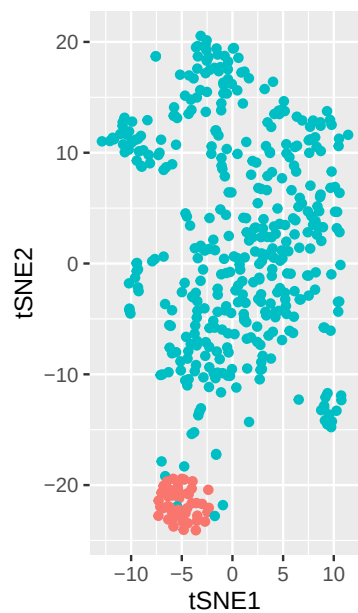
We calculate the variance of each principal component and the proportion of the overall variance that the components are explaining.

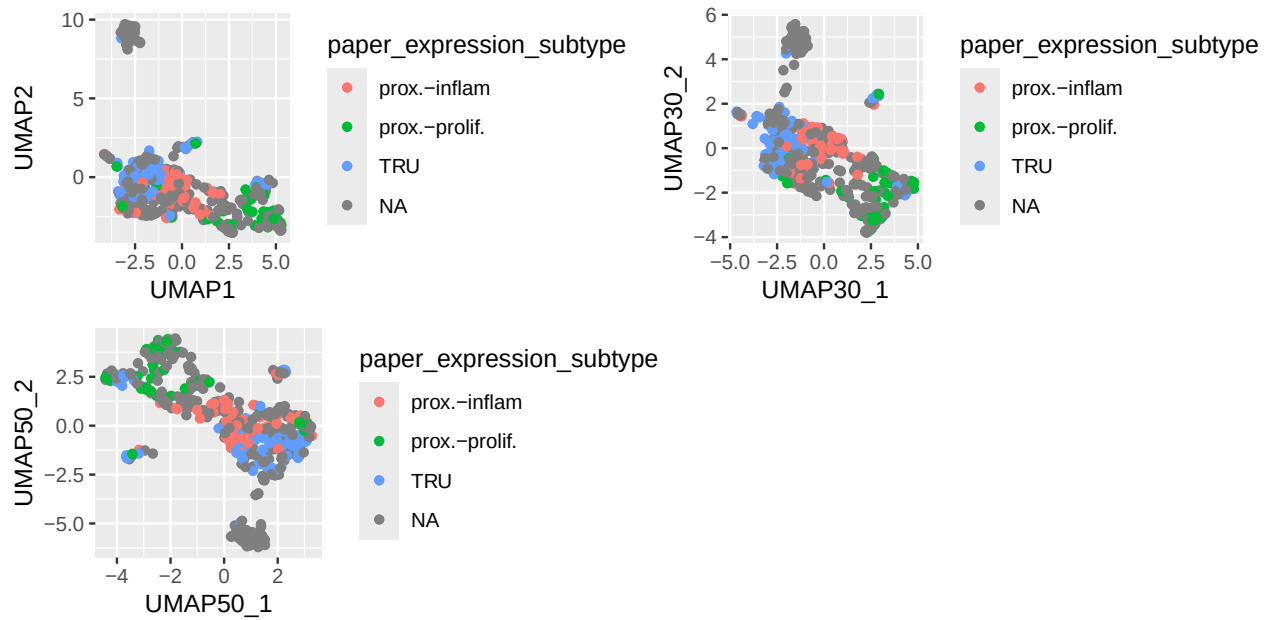


The tumor samples are clustered together and away from the normal samples.



PCA isn't separating the feature subtypes clearly so we perform t-distributed Stochastic Neighbor Embedding. We first want to see again if the normal and tumor samples are separating, after we separate the expression subtypes. They are roughly clustering together.

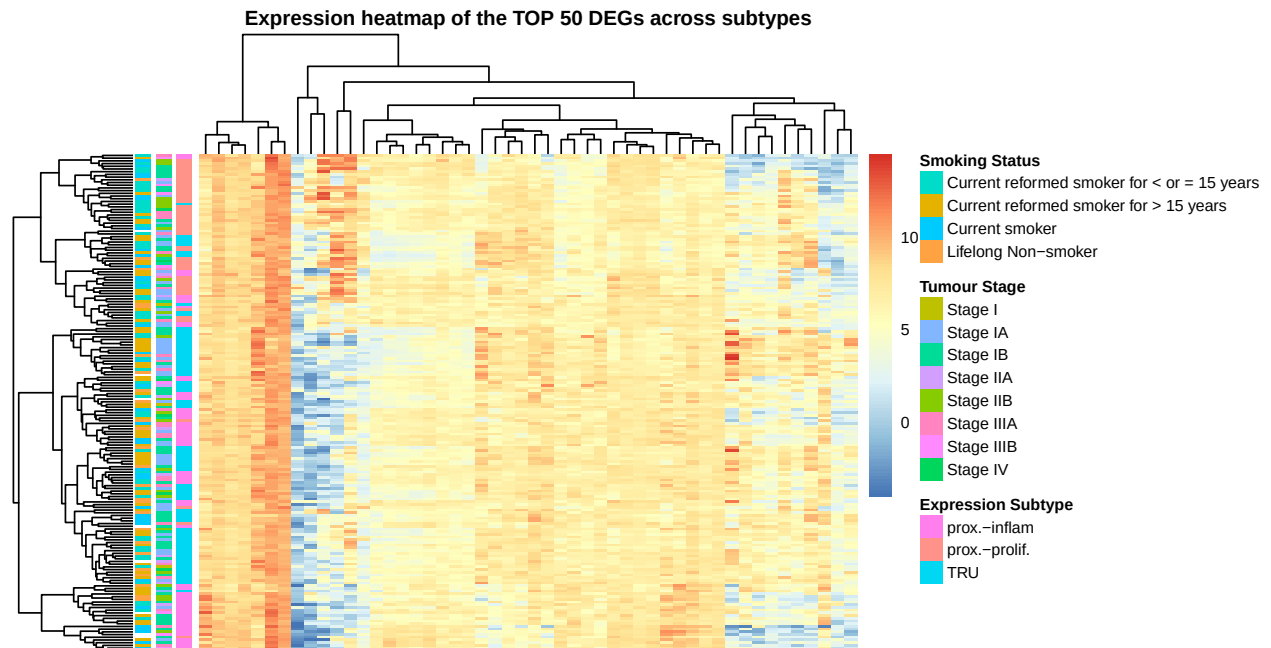




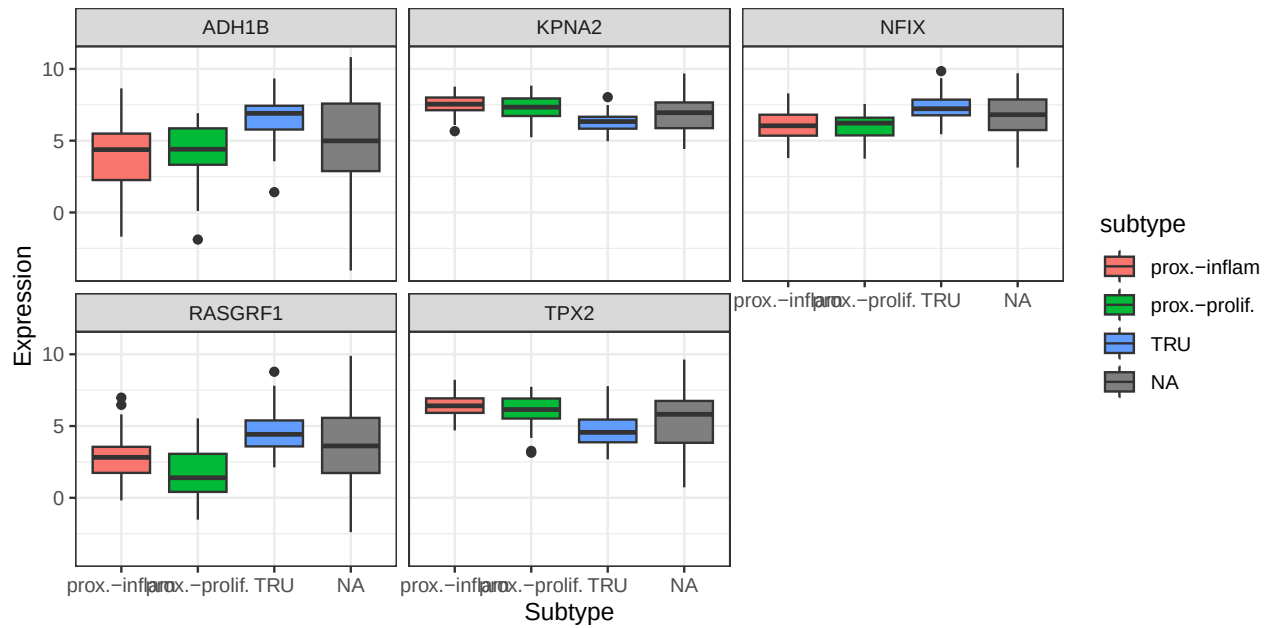
With only the 10 top PCs the clusters are more clearly clustered together. More PCs bring more noise to the UMAP plot.

Differential Expression Analysis

We now set out to perform differential expression analysis between the different tumour subtypes. After Benjamini-Hochberg correction, we visualise the Top 50 most significant differentially expressed genes in a heatmap.

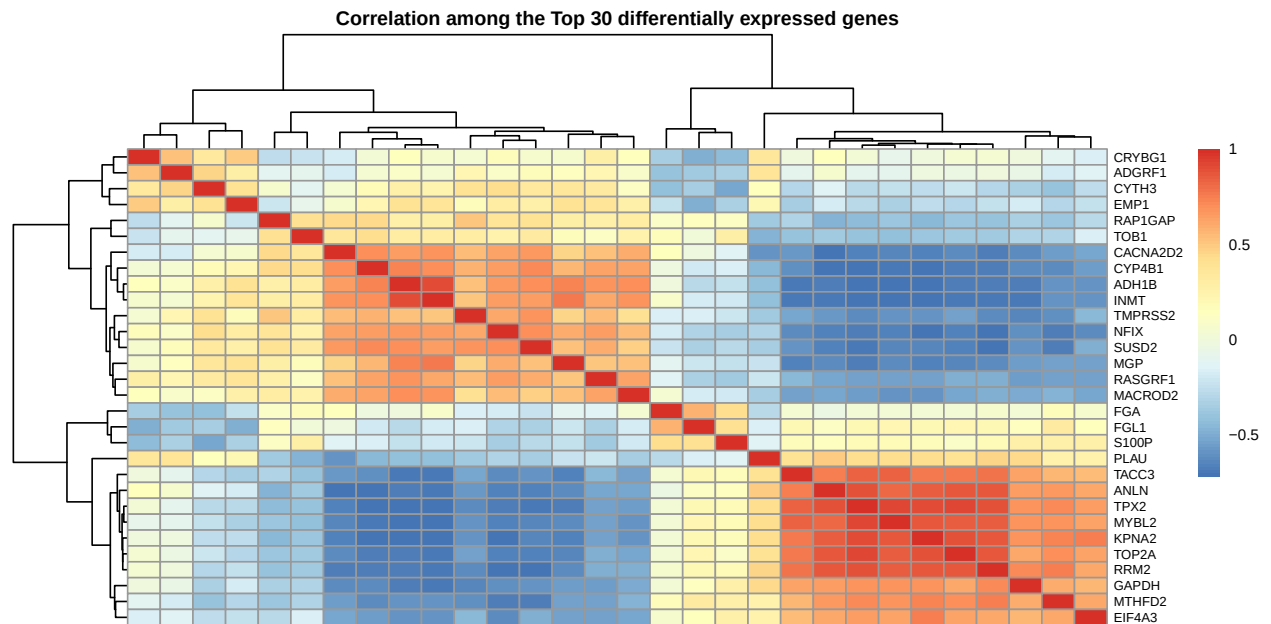


We then compare the top 5 DEGs using a boxplot. TRU seems to be the more clearly differentiated subtype compared to prox.-inflam. and prox.-prolif.. Most of the genes are well known in tumour research, namely KPNA2, NFIX, RASGRF1 and, TPX2. ADH1B is less strongly associated with tumours, yet some finding do link it to reduced tumour stemness.

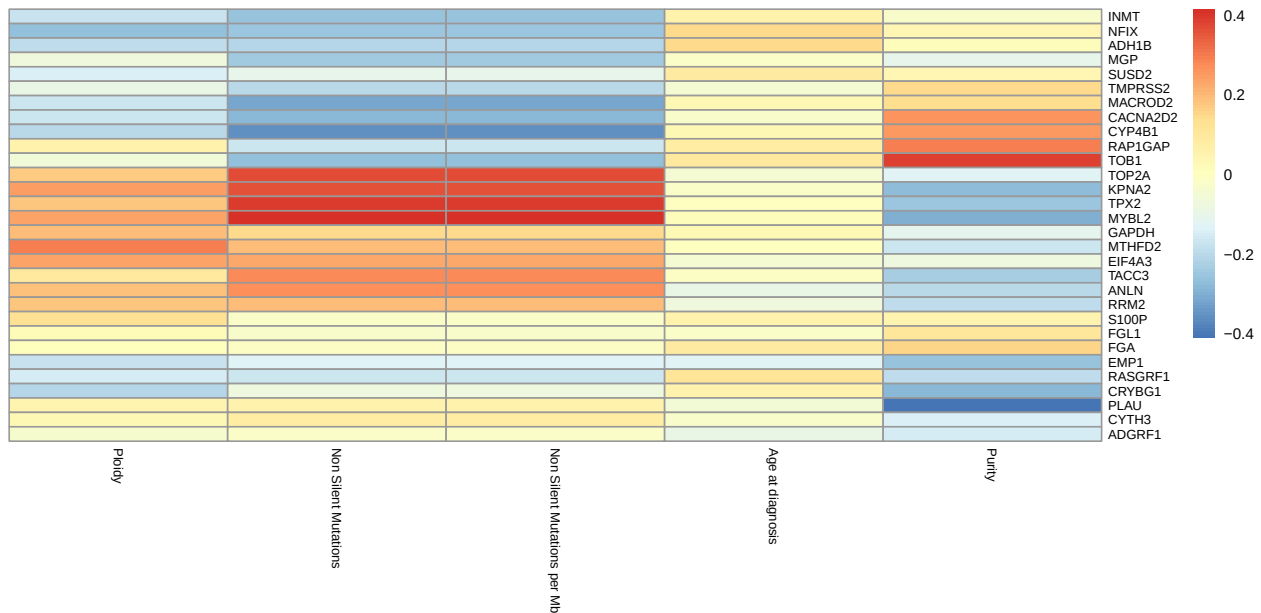


Expression Correlation amongst Highly Differentially Expressed Genes

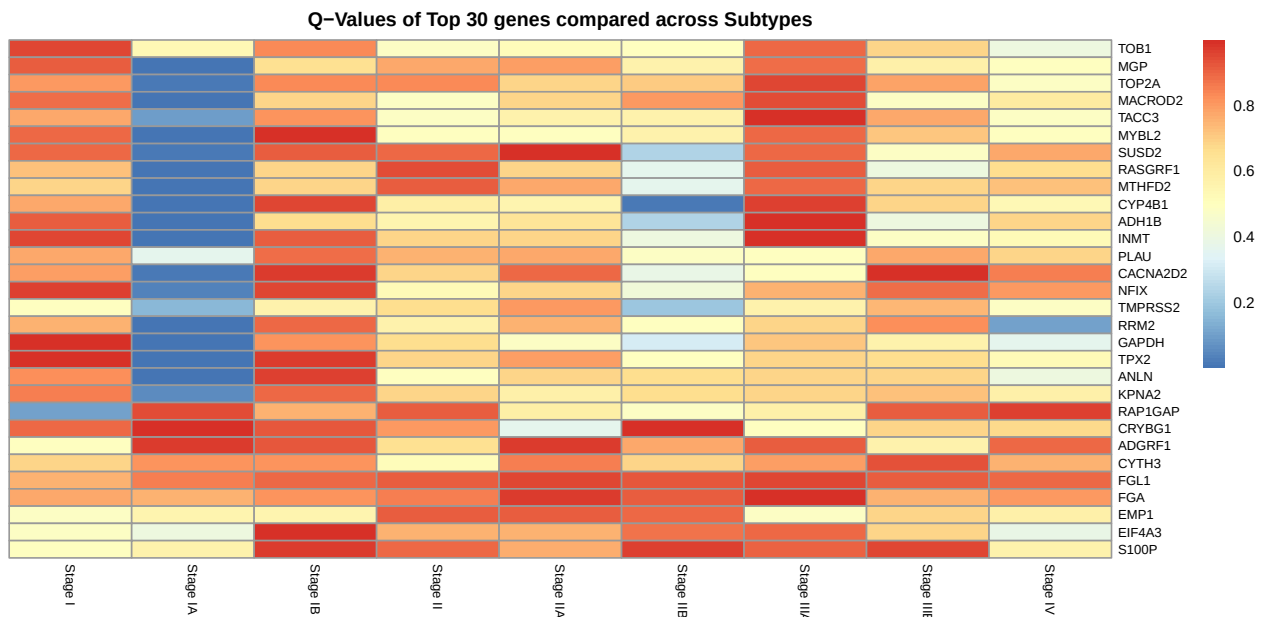
We can also compare the correlation between top DEGs to find co-expressed or anti-expressed modules. There seems to be some genes which are in co-expressed and a separate module which seems inversely correlated.



Furthermore, we want to compare the top 30 DEGs to clinical feature. For this, we compute the correlation between continous clincial variables and the top 30 DEGs.



Finally, with respect to the differential gene expression, we want to ascertain how top 30 DEG expression differs between tumour stages. Curiously, many of the top DEGs are differentially expressed only in Stage IA vs. rest



Predictive Modelling

Based on Expression Data

Target: Expression Subtype

First we initialise a mutable combined data set from our immutable annot and exp data sets, discard unnecessary annotation columns, and split the data into a training and testing data set. We also scale and centre both data sets based on the training data.

Using the scaled and centred training data, we then train random forest and multinomial logistical regression models and make a prediction each.

We want to take a look at the 5 most important variables for both models. For the random forest regression, we simply use variable importance as the metric. For multinomial logistical regression, we use the euclidean deviance of the coefficients from zero for each gene. These differ, though there is overlap.

We then walk through both lists of genes sorted by their importance to the models until we reach an overlapping selection of 5. The given depth shows us how many of the sorted list we need to include to get our overlap.

Target: Tumor Stage

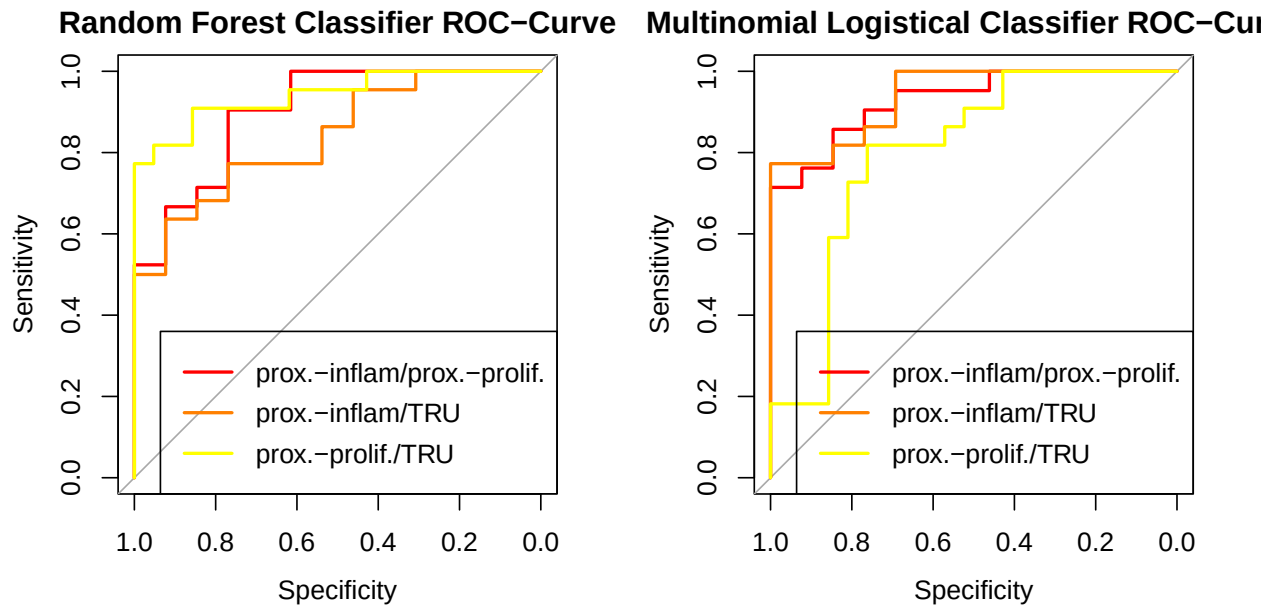
We now do the exact same for the prediction of tumor stage, with the addition of splitting the tumor stages into only two categories: Early and Late.

If we look at the importance of the various genes to the prediction, we again see some overlap between the random forest and binomial logistical regressions in the top 5 most important variables.

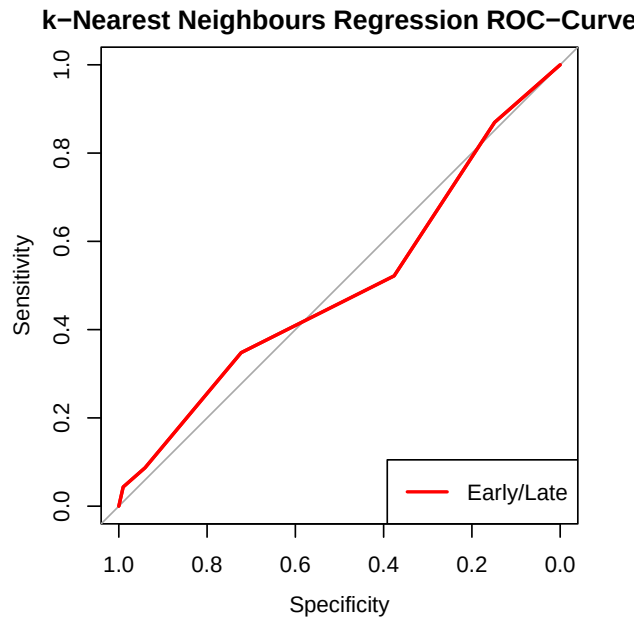
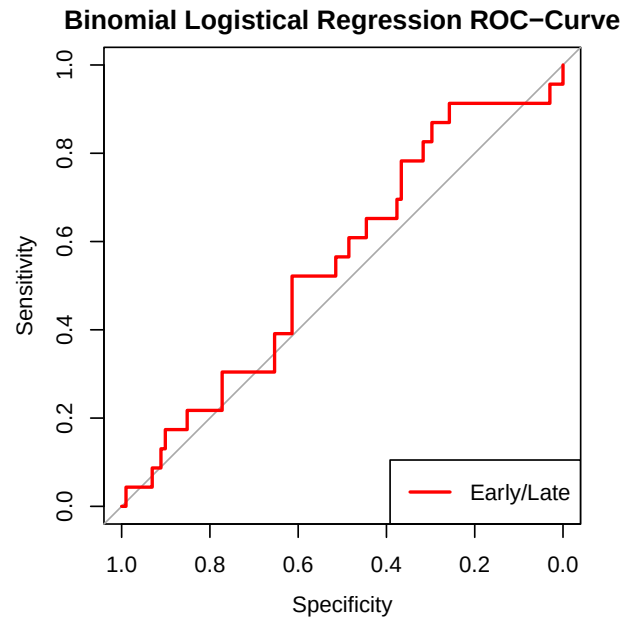
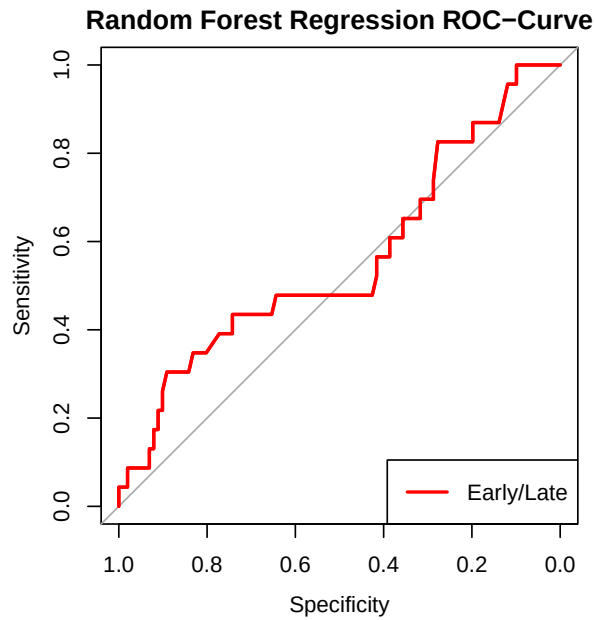
We thus again look at how far we need to go to find an overlap between all three models of 5 important variables.

Quality and Comparison of Models

To compare the quality of the models, we take a look at ROC curves to ascertain the quality of the expression subtype classifiers. The random forest classifier outstrips the multinomial logistical one, especially when it comes to distinguishing proximal-inflammatory from proximal-proliferative biopsies.



We now take a look at the ROC curves for the tumour stage classifiers. We see that random forest classification and kNN outstrip logistical regression.



When we look at key values for these classifiers (first table expression subtype classification, second table tumour stage classification) we can see why some of them don't perform well.

##	Random.Forest	Multinomial.Logistical
## Accuracy	0.6250000	0.6607143
## Macro-Averaged F1	0.6175156	0.6601431
## Precision for Class: prox.-inflam	0.5000000	0.5789474
## Precision for Class: prox.-prolif.	0.7500000	0.7333333
## Precision for Class: TRU	0.6250000	0.6818182
## Recall for Class: prox.-inflam	0.6153846	0.8461538
## Recall for Class: prox.-prolif.	0.5714286	0.5238095
## Recall for Class: TRU	0.6818182	0.6818182

##	Random.Forest	Binomial.Logistical	k-Nearest-Neighbours
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## Accuracy	0.8145161	0.8064516	0.8145161
## F1	0.8977778	0.8918919	0.8968610
## Precision	0.8145161	0.8181818	0.8196721
## Recall	1.0000000	0.9801980	0.9900990

In the prediction of tumour stage, we have the problem, that our samples at Early stage far outnumber those at Late stage.

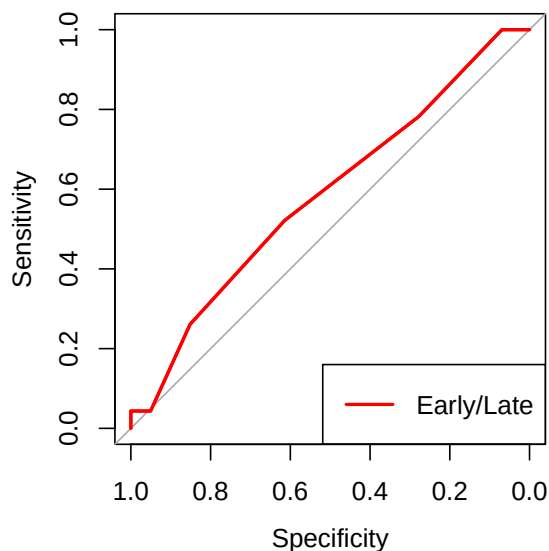
```
##
## Early Late
## 225 62
```

Rerunning training for the best performer (highest accuracy and precision) using a down-sampled training data set significantly worsens performance, though precision improves somewhat.

## Accuracy	F1	Precision	Recall
## 0.5967742	0.7126437	0.8493151	0.6138614

Also the ROC Curve looks less healthy.

ROC-Curve of k-Nearest Neighbours Regre:



We determined the top 5 genes for the classifiers for each of the classification problems (expression subtype, tumour stage) by looking at the best-performer and by integrating across classifiers to find an overlap of top-5 important genes. We can here see, that the top-5 differentially expressed genes have no overlap with the most important predictors.

This is somewhat surprising, but may be due to the DEGs being found by 1-vs-rest comparison and the classifiers lumping all samples of a expression subtype or tumour stage together. Furthermore, we may find overlap, when looking at genes less important to the classifiers. We do so for one example.

To find an overlap, we need to go to a depth of 43, which is fascinating, but probably due DEGs being determined by difference in 1 group vs the rest, whereas the classifiers use simple the top 50 genes with the highest variance, irrespective of group. We sadly don't have page space to show a rerun of the classifiers for expression subtype.

The top-5 most important genes of our best classifier for the expression subtype (random forest) are: - FGL1: binds to inhibitory receptor LAG-3 on activated T-cells, involved in T-cell modulation, and immune evasion by cancer cells (Xu *et al.*, 2026). - PGC: role in regulation of mitochondrial biogenesis, improves mitochondrial capacity of cancer cells (Akasha *et al.*, 2025). - FGA: inhibited by TGF- in context of high IL-6 expression, common in lung cancer (Han *et al.*, 2024; Schafer *et al.*, 2007). - AQP5: restores cancer cell proliferation in colorectal cancer cells under 5-FU

exposure, knockdown inhibits tumorigenesis in and suppresses proliferation, migration, and self-renewal of gastric cancer stem cells (Yang *et al.*, 2025; Gao *et al.*, 2024). - LGALS4: inhibits glycolysis and promotes apoptosis in colorectal cancer cells, down regulation would be beneficial to cancer cells (Li *et al.*, 2025).

All of these have significance to cancers. Depending on their regulation, they can modulate cancer cell proliferation, survival, and aggressiveness.

Based on Clinical Annotation

we remove all rows with NAs

prepare expression_subtype data, convert names to valid R variable names

convert the categorical values into numerical ones

divide in train and test data 70/30

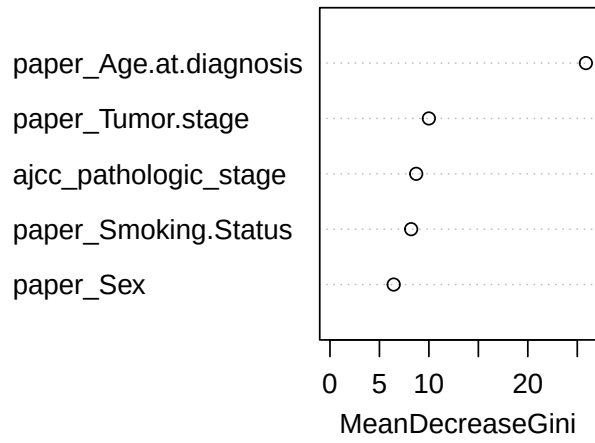
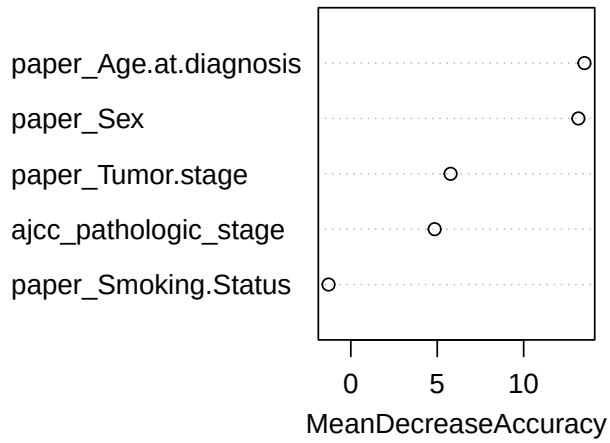
calculate a multinomial logistic regression (was too complex with all features so we chose 3: Age at diagnosis, Tumor.stage, Smoking.status) We tried different combinations of features to get the best possible model. The model with these 3 features makes the most sense to predict expression_subtype (model has the highest accuracy)

predicting with the test data

```
## Confusion Matrix and Statistics
##
##           Reference
## Prediction PI PP TRU
##           PI    4  4   1
##           PP    1  4   2
##           TRU  10  4  17
##
## Overall Statistics
##
##           Accuracy : 0.5319
##           95% CI : (0.3808, 0.6789)
##           No Information Rate : 0.4255
##           P-Value [Acc > NIR] : 0.09277
##
##           Kappa : 0.2453
##
## Mcnemar's Test P-Value : 0.02007
##
## Statistics by Class:
##
##           Class: PI Class: PP Class: TRU
## Sensitivity          0.26667  0.33333  0.8500
## Specificity          0.84375  0.91429  0.4815
## Pos Pred Value       0.44444  0.57143  0.5484
## Neg Pred Value       0.71053  0.80000  0.8125
## Prevalence           0.31915  0.25532  0.4255
## Detection Rate       0.08511  0.08511  0.3617
## Detection Prevalence 0.19149  0.14894  0.6596
## Balanced Accuracy    0.55521  0.62381  0.6657
```

Random Forest

Feature Importance



```
## Confusion Matrix and Statistics
##
##           Reference
## Prediction PI PP TRU
##           PI   4  4  4
##           PP   2  4  4
##           TRU  9  4 12
##
## Overall Statistics
##
##           Accuracy : 0.4255
##           95% CI : (0.2826, 0.5782)
##           No Information Rate : 0.4255
##           P-Value [Acc > NIR] : 0.5557
##
##           Kappa : 0.0994
##
## Mcnemar's Test P-Value : 0.4593
##
## Statistics by Class:
##
##           Class: PI Class: PP Class: TRU
## Sensitivity      0.26667  0.33333  0.6000
## Specificity      0.75000  0.82857  0.5185
## Pos Pred Value   0.33333  0.40000  0.4800
## Neg Pred Value   0.68571  0.78378  0.6364
## Prevalence       0.31915  0.25532  0.4255
## Detection Rate   0.08511  0.08511  0.2553
## Detection Prevalence 0.25532  0.21277  0.5319
## Balanced Accuracy 0.50833  0.58095  0.5593
```

k-nearest neighbors

First we normalize the data (only features not the outcome, expression_subtype is in column 5 so we are excluding it)

train the model

testing the knn model

```

## Confusion Matrix and Statistics
##
##           Reference
## Prediction PI PP TRU
##           PI   2   2   1
##           PP   0   3   1
##           TRU 13   7  18
##
## Overall Statistics
##
##           Accuracy : 0.4894
##           95% CI : (0.3408, 0.6394)
##           No Information Rate : 0.4255
##           P-Value [Acc > NIR] : 0.2295480
##
##           Kappa : 0.1493
##
## Mcnemar's Test P-Value : 0.0007822
##
## Statistics by Class:
##
##           Class: PI Class: PP Class: TRU
## Sensitivity           0.13333  0.25000  0.9000
## Specificity           0.90625  0.97143  0.2593
## Pos Pred Value        0.40000  0.75000  0.4737
## Neg Pred Value        0.69048  0.79070  0.7778
## Prevalence            0.31915  0.25532  0.4255
## Detection Rate        0.04255  0.06383  0.3830
## Detection Prevalence  0.10638  0.08511  0.8085
## Balanced Accuracy      0.51979  0.61071  0.5796

```

AUC ROC for all 3 models

```
## Multi-class area under the curve: 0.607
```

```
## Multi-class area under the curve: 0.7639
```

```
## Multi-class area under the curve: 0.5
```

The logistic regression model We are comparing all 3 models for predicting the variables “PI”, “PP” and “TRU” seperatly by plotting the ROC curves. We see that the logistic regression model is performing better than the random forest.

```
## Setting levels: control = FALSE, case = TRUE
```

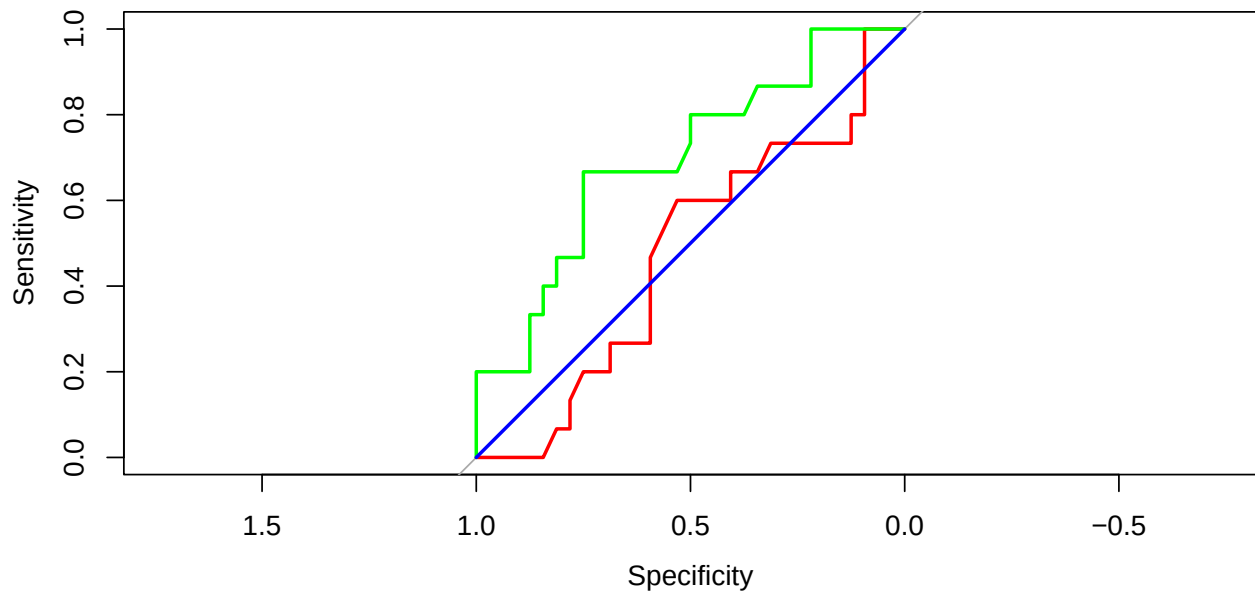
```
## Setting direction: controls > cases
```

```
## Setting levels: control = FALSE, case = TRUE
```

```
## Setting direction: controls < cases
```

```
## Setting levels: control = FALSE, case = TRUE
```

```
## Setting direction: controls < cases
```



```
## Setting levels: control = FALSE, case = TRUE
```

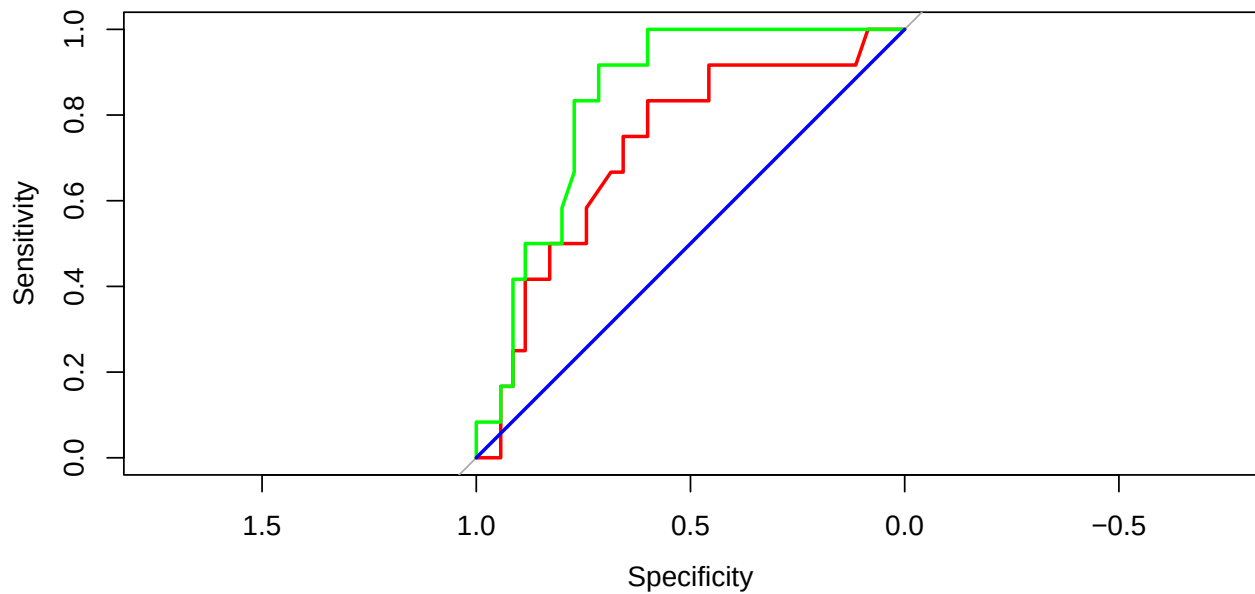
```
## Setting direction: controls < cases
```

```
## Setting levels: control = FALSE, case = TRUE
```

```
## Setting direction: controls < cases
```

```
## Setting levels: control = FALSE, case = TRUE
```

```
## Setting direction: controls < cases
```



```
## Setting levels: control = FALSE, case = TRUE
```

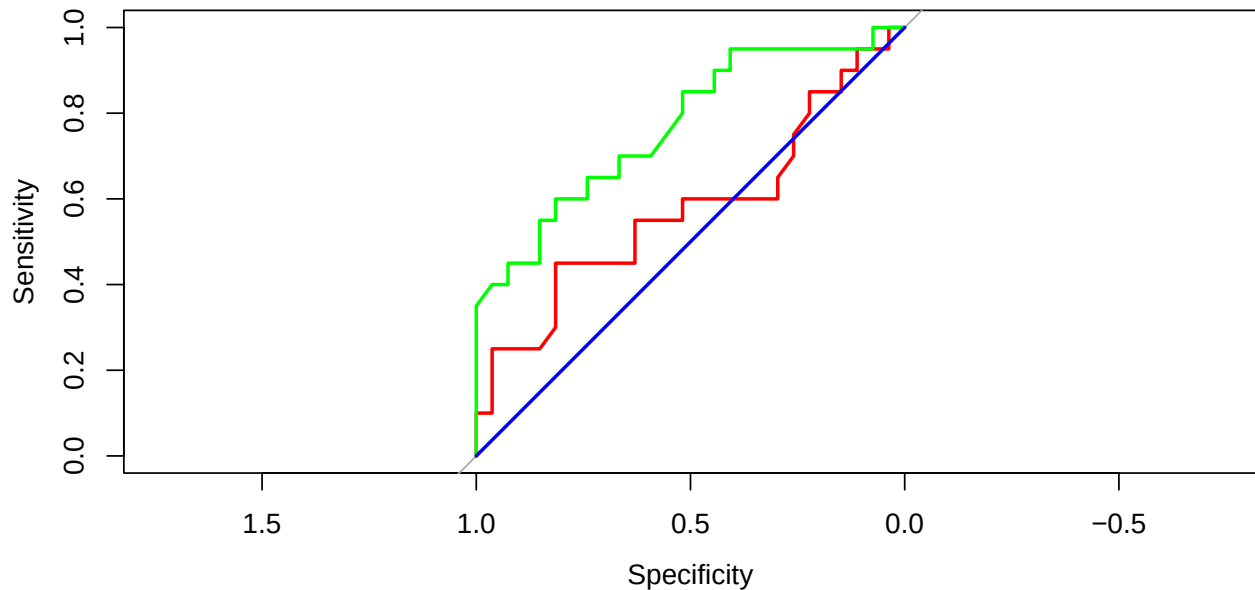
```
## Setting direction: controls < cases

## Setting levels: control = FALSE, case = TRUE

## Setting direction: controls < cases

## Setting levels: control = FALSE, case = TRUE

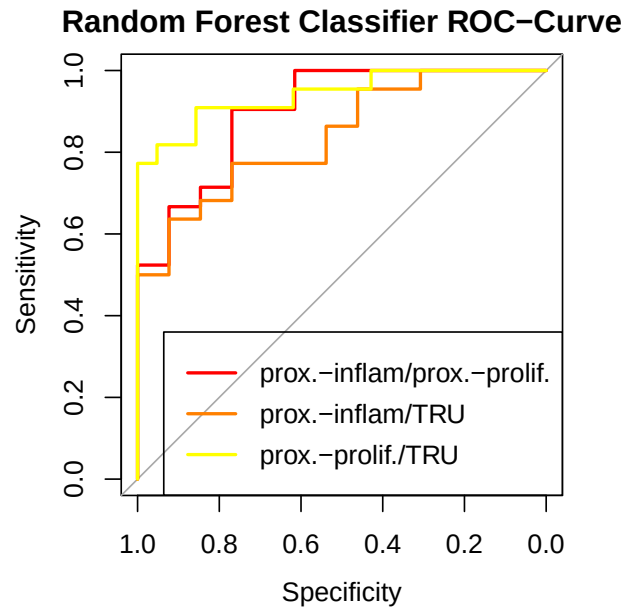
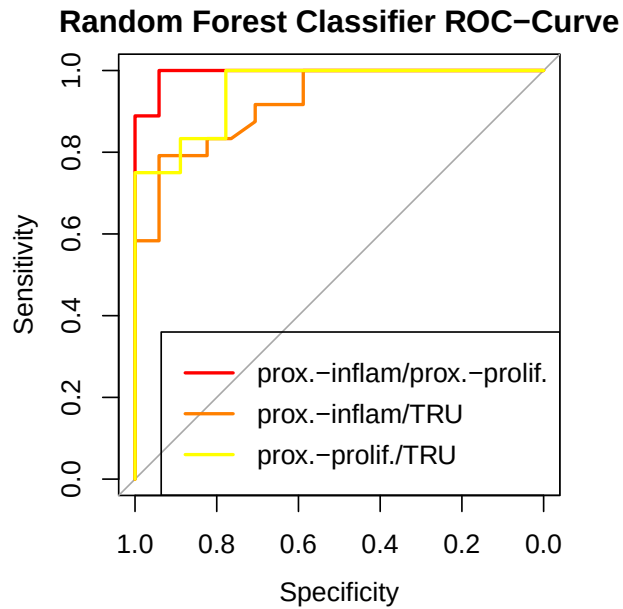
## Setting direction: controls < cases
```



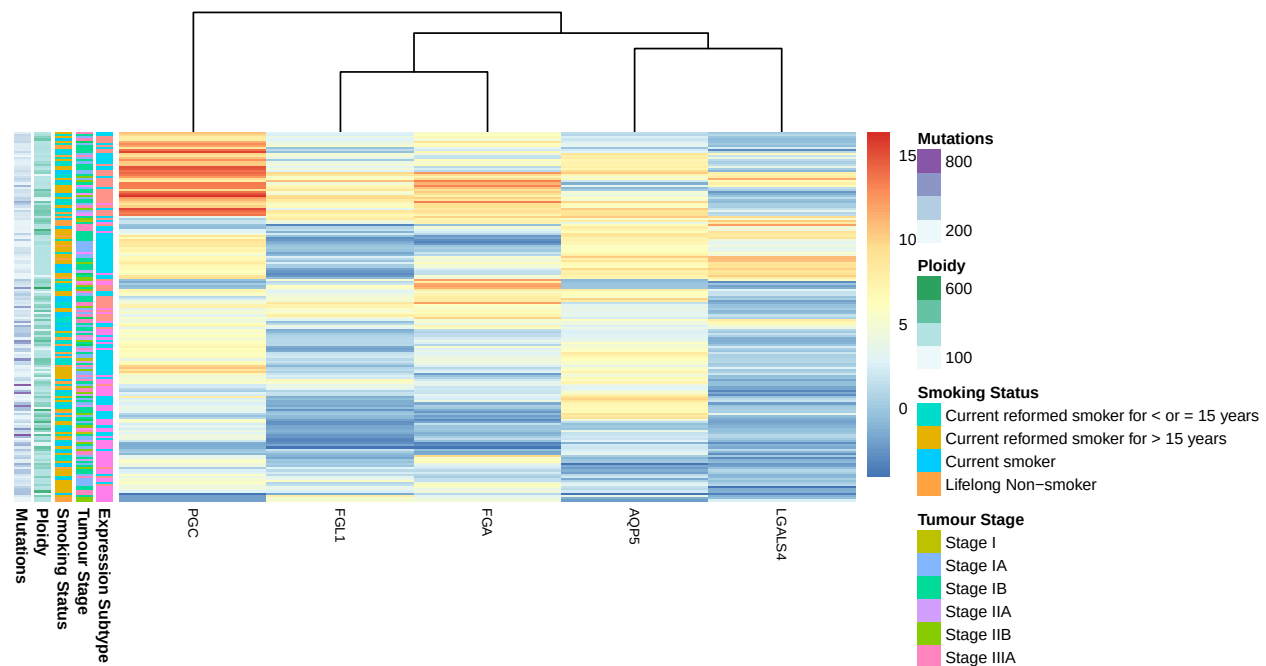
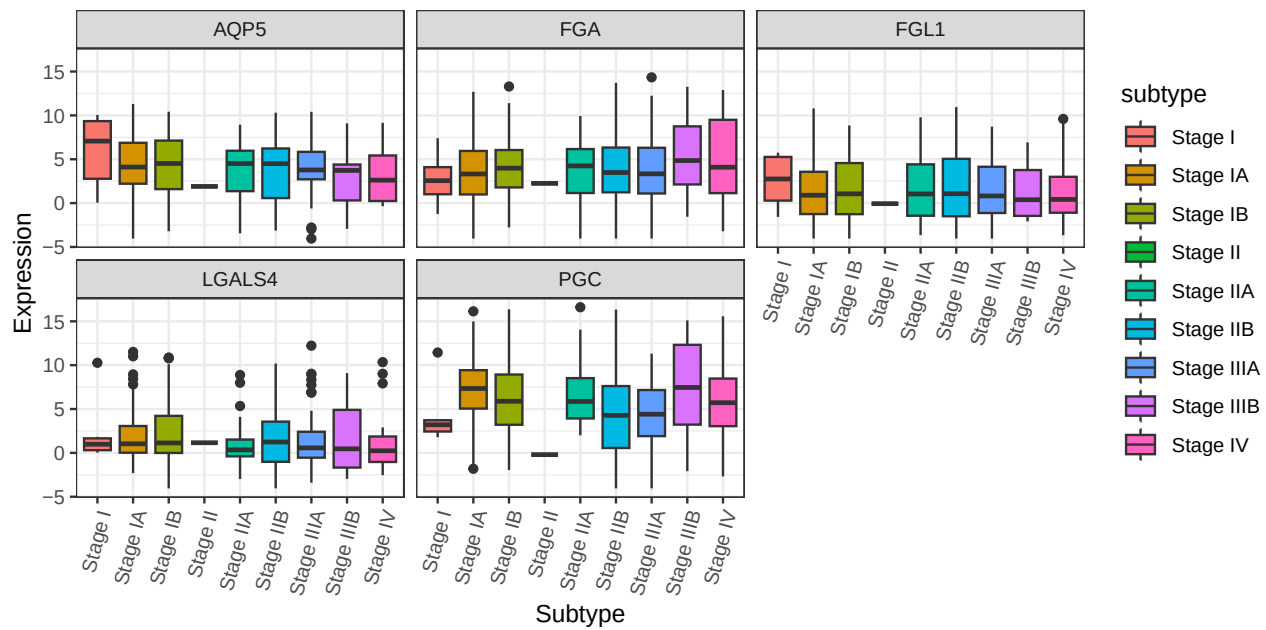
The knn model showed poor predictive performance with an AUC of 0.5, which means that the prediction it makes is not better than guessing by chance. We know that k nearest neighbor is a distance based model, that's why it is not performing well with categorical variables. We tried to encode dummy variables, to have numerical values but it didn't work.

Integrative Classifier

We now build a integrative classifier, using the top-25 most important genes for our random forest classifier targeting expression subtype as well as the clinical variables "expression_subtype", "Tumor.stage", "Nonsilent.Mutations", "Ploidy.ABSOLUTE.calls", "Purity.ABSOLUTE.calls", "Smoking.Status", and "Age.at.diagnosis". As can be seen from the two ROC graphs and the comparison table of key characteristics, this doesn't improve the model, despite the training data having almost the same number of observations.



```
## Integrated.Random.Forest
## Accuracy 0.8000000
## Macro-Averaged F1 0.7975368
## Precision for Class: prox.-inflam 0.7619048
## Precision for Class: prox.-prolif. 0.7777778
## Precision for Class: TRU 0.8500000
## Recall for Class: prox.-inflam 0.9411765
## Recall for Class: prox.-prolif. 0.7777778
## Recall for Class: TRU 0.7083333
## Non.integrated.Random.Forest
## Accuracy 0.6250000
## Macro-Averaged F1 0.6175156
## Precision for Class: prox.-inflam 0.5000000
## Precision for Class: prox.-prolif. 0.7500000
## Precision for Class: TRU 0.6250000
## Recall for Class: prox.-inflam 0.6153846
## Recall for Class: prox.-prolif. 0.5714286
## Recall for Class: TRU 0.6818182
```



Discussion

References

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